

Differential diagnoses for shoulder injuries include: biceps tendon injury, triceps muscle, pectoralis muscle, infraspinatus, supraspinatus, or other cuff muscle injury, medial or lateral glenohumeral ligament damage, trauma to the scapula, proximal humerus, or entire joint including luxation, degenerative joint disease, osteochondrosis dissecans, panosteitis, joint infection, or arthritis.

Injuries to the biceps are among the most commonly seen orthopedic shoulder injuries in the racing sled dog. Injuries to the triceps muscle or the pectoral muscles can also be observed frequently. In general triceps and pectoral injuries can more often be massaged and treated on the trail and dogs can finish the race, when compared to biceps muscle injuries. Shoulder injuries may be more frequently observed in speed racing sled dogs, but are also commonly seen during long distance races. The authors have noticed these injuries to occur frequently in dogs that covered long distances running on ice, or after high-stepping through deep snow. Injured dogs show a gait with the head going down when placing the healthy foot on the ground, and moving the head up when placing the injured leg on the ground ("head-bob", also see the chapter on orthopedic examination of the sled dog). Dogs with biceps injuries are painful upon extension of the shoulder while simultaneously applying pressure on the tendon within the bicipital groove. Triceps injuries are diagnosed by eliciting a painful response when extending the leg, and when deeply palpating the different heads of the triceps muscle. In severe cases, triceps avulsion injuries from the olecranon can be seen. These dogs show pronounced swelling just proximal to the olecranon, and are painful upon palpation of this area. Treatment for biceps or triceps injuries include massages, application of warm packs (Figure 10), and possibly more sophisticated treatments as described in the section of "general treatment for muscle injuries".



Figure 10: Warm pack to prevent or treat shoulder injuries.

It may be helpful to remove the dog from the team and walk it for 10-20 minutes prior to leaving the checkpoint. This will help the muscle to warm up, and possibly prevent more damage. Some mushers perform massages even when on the trail at various intervals. Once a dog had a muscle injury the treatment should be automatically performed at every following checkpoint. Dogs with biceps or triceps injuries may sometimes "warm out" of the injury, and may finish the race. However, most dogs with injuries to these structures will have to be dropped. Some mushers will drop dogs with injured shoulders as soon as they are aware of this cause for lameness to minimize possible long-term damage. The key for a good recovery is appropriate treatment for muscle injuries, rest, and appropriate rehabilitation. Dedicated protocols for physical rehabilitation to treat shoulder injuries are available, and the application of a specifically designed pair of shoulder hobbles may improve outcome.⁷ Repetitive injuries of the biceps tendon, the infra- or supraspinatus tendon, or the glenohumeral ligaments may result in chronic damage. In these cases it may be necessary to apply advanced treatment methods, such as arthroscopy, and / or surgical repair of the damaged tissues, for example placement of prosthetic ligaments in cases of shoulder instability. The recovery time for shoulder injuries can be as long as 3-6 months. Shoulder injuries may have a significantly better chance for rapid recovery if diagnosed within 12 hours of occurrence, if non-steroidal anti-inflammatory is administered within 24 hours, if the dog receives a thorough, deep massage (see above) of all shoulder muscles, and if the dog is confined to a small space for 48 hours.⁸ If surgery may be necessary the use of arthroscopy may allow a quick diagnosis and possible simultaneous treatment during one session.

CARPAL INJURIES

The diagnosis of carpal injuries is made by an appropriate orthopedic examination (see previous chapter).

Differential diagnoses for carpal injuries include: Severe exercise induced joint inflammation, complete or partial luxation, cartilage damage to the joint surface, fissures or fractures of the carpal bones, severe ligamentous injuries, such as hyperextension damage, or degenerative joint disease.

Carpal injuries are common during the first longer training runs in the season or during the first 1-500 miles during long distance races. These injuries are commonly observed when the trail is hard, or leads through rough, hilly terrain. Animals with severe discomfort will show the typical “head-bob”, typical for front leg lameness. Most wrist injuries arise secondary to heavy exercise and present as marked swelling of the wrist from significant inflammation of the joint. Repetitive high impact trauma leads to increased blood flow and production of inflammatory joint fluid. The joint capsules of the carpus are tight, and an increase in joint fluid leads quickly to increased pressure on pain receptors within the joint capsule, possibly leading to clinical signs of lameness. The joint should be thoroughly massaged at the checkpoint, and range of motion exercises should be performed. The carpus is flexed and extended several times. Furthermore, the application of joint health supporting liniments or ointments (see above) in combination with some cellophane wrap and a custom made wrist wrap, or “sweat” wrap (neoprene and Velcro™ straps) seem to frequently improve the condition (Figure 11). Wraps may also be applied as a preventative measure at every checkpoint to interrupt the mechanism for joint swelling before it begins. The wraps are left in place for the duration of rest at the specific checkpoint. It is imperative not to place the wrap too tight and to cover the leg completely all the way to the toes in order to prevent venous obstruction. If the dog’s foot swells up after application of the bandage, it was too tight! A dog must not run with the wrist wrap on! The application of wrist wraps in combination with an ointment may help reduce the risk for injury. However, appropriate conditioning and slowing down on hills may be more helpful in preventing wrist injuries from occurring. Furthermore, some mushers believe that there is a relation between the degree of hydration status of the animal and the occurrence of carpal injuries. There is no scientific evidence for this statement, but it is always recommended to maintain the best possible hydration status in racing sled dogs to decrease the risk of any exercise related disease or injury. Most sled dogs with carpal swelling improve throughout the course of the race. On the other hand, repeated or prolonged trauma, cartilage wear and development of arthritis may result in consistently decreased range of motion and early onset of lameness after running comparably short distances. The prognosis for chronic carpal injuries in the sled dog is guarded. With chronic lameness, arthrodesis (joint fusion) may be the only way to provide the animal with pain free ambulation. Fractures, luxations, or ligamentous injuries frequently need to be treated by surgical stabilization, such as arthrodesis (Figure 12). Dogs with fused carpi can often return to an athletic life style. The specific form of treatment and the prognosis depend on the underlying condition and will have to be discussed with an appropriately trained veterinarian or veterinary surgeon.



Figure 11: Wrist wraps often help to overcome swollen carpi.

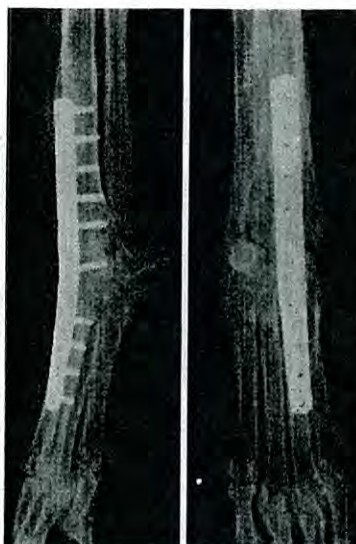


Figure 12: Carpal arthrodesis may be necessary to treat severe injuries of the carpus. Dogs with fused carpi can often return to an athletic life style.

ELBOW INJURIES

The diagnosis of elbow injuries is made by an appropriate orthopedic examination (see previous chapter). Differentials for lameness arising from an elbow problem include: Fracture, luxation, infection, arthritis, ununited coronoid process, ununited anconeal process, incomplete ossification of the humeral condyles, ununited medial epicondyle, osteochondrosis dissecans, or elbow incongruency. Most of these diseases require a thorough diagnostic work up (radiographs, computed tomography, arthroscopy). Treatment consists mostly of supportive care and/or surgery. Sled dogs with elbow problems that arise during a race are unlikely to finish the event. Depending on the underlying disease, the prognosis after surgical treatment varies, depending on the underlying disease.

HIP INJURIES

The diagnosis of hip injuries is made by an appropriate orthopedic examination (see previous chapter). The most common cause of hip injuries in the sled dog is muscle strain. Psoas muscle injuries can be seen and may be confused with injuries to the hip joint. The psoas muscles consist of a group of deep stomach muscles attaching at the pelvis, femur and ventral lumbar spine. It is not possible to palpate most of these muscles, because of their deep location. This muscle group is contracted to bring the hind legs underneath the body when running, resulting in forward propulsion. Contraction of these muscles arches the lumbar spine and changes the angle of the long axis of the pelvis to a steeper slope. This movement may be compared to winding up a spring for an explosive release. If this muscle group is injured, the dog may present with a pronounced upward bend of the lower back. The dog shows decreased ability of hip extension. Severe injury of these muscles may result in uncomfortable gait for several months. Appropriate conditioning and stretching at least every other day may prevent this injury from occurring. Muscle strains to the hip may be treated using conventional methods. Massage, warm packs, and careful stretching may help to improve the condition. Similar treatment can be applied to other muscle injuries of the hip joint. More information on muscle injury is provided in the section of "general treatment for muscle injuries". Administration of non-steroidal anti-inflammatories, muscle relaxants such as methocarbamol or application of TENS (see above) may also help.

Differentials for surgical problems of the hip include: Fracture, luxation, and degenerative changes such as arthritis or hip dysplasia. Treatment and prognosis depend on the underlying condition. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities.

STIFLE INJURIES

The diagnosis of stifle injuries is made by an appropriate orthopedic examination (see previous chapter). Most stifle injuries are secondary to trauma or advanced degenerative changes, and conservative treatment is unlikely to return the animal to racing conditions. Cranial cruciate ligament rupture (CCLR) is the most common orthopedic injury of the dog. History, signalment, gait, and orthopedic examination usually provide good information on the specific stifle injury. To confirm the diagnosis, radiographs are frequently taken. Further diagnostic imaging may include arthroscopy (Figure 13) or magnetic resonance imaging.

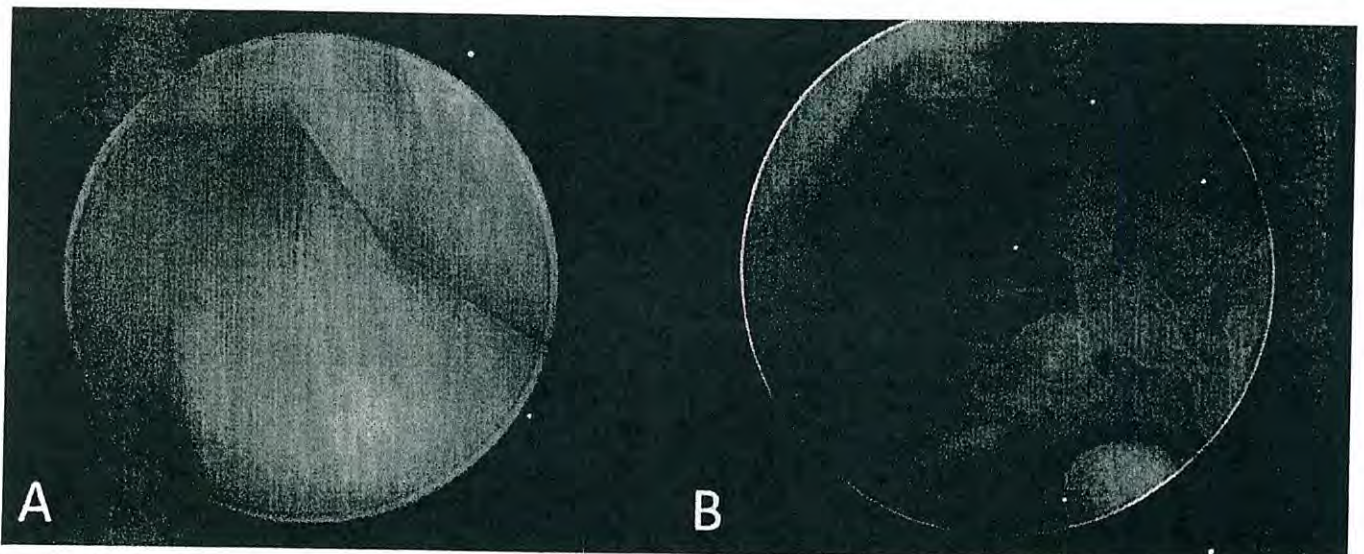


Figure 13: Arthroscopy of the stifle. Intact cranial cruciate ligament (A), severe joint inflammation secondary to ruptured cranial cruciate ligament (B). (Pictures: Brian Beale, DVM, MS, DACVS, Gulf Coast Clinic, Houston).

The surgical procedure of choice for CCLR in the first author's hands is a Tibial Plateau Leveling Osteotomy (TPLO). In a nutshell, with this procedure the direction of forces within the stifle joint is altered, resulting in a stable knee during weight bearing. There is a high likelihood for operated animals to return to racing activity if the procedure is performed accurately, and once healing is completed (Figure 14). To achieve best results, it is highly recommended to follow a strict physical rehabilitation protocol during the recovery period.

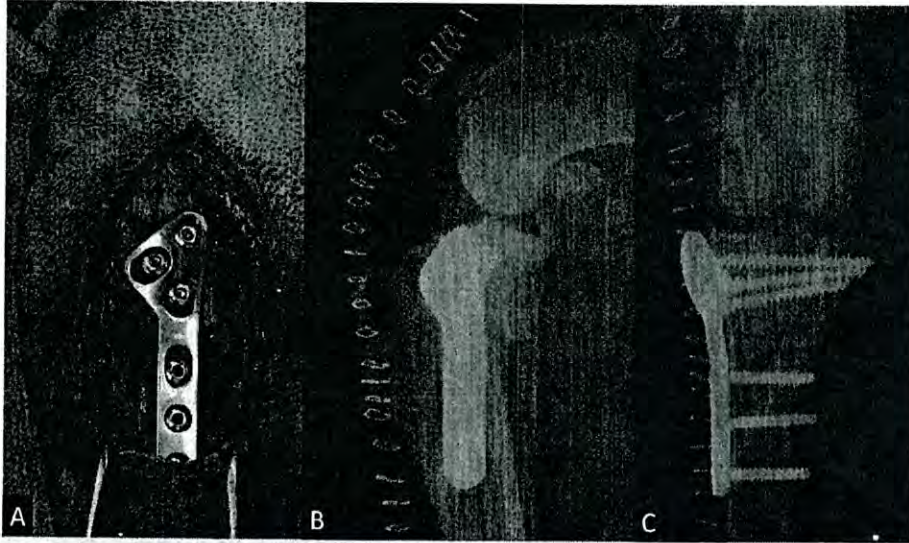


Figure 14: TPLO provides usually excellent return to function after a cranial cruciate ligament rupture. During surgery (A), post-operative radiographs lateral view (B) and anterior-posterior view (C).

Patella luxations (PL) are less likely to be a cause for lameness in a racing sled dog. However, it is a common orthopedic condition in the dog, and may be the cause for lameness in a young dog that just started training. Treatment is surgical and the prognosis is good, if the procedure is performed appropriately. To determine the best treatment for PL, several factors have to be taken into consideration. A grading system is described based on the amount of PL. Grade I: Luxation occurs only intermittently. The patella can be luxated manually, but replaces itself immediately. Grade II: After manual luxation, the stifle is flexed and extended, and the patella remains luxated. Grade III: The patella is luxated at all times, but can be pushed into normal anatomical position. Without manipulation of the stifle joint, the patella returns to the outside of the knee joint. Grade IV: The patella is luxated permanently and cannot be reduced manually. PL can occur medially or laterally. Most small breeds have medial luxations, most large breeds lateral luxations. Lateral luxations seem to be more common in dogs with straight legs and high riding patellas (Akita, Malamute). The treatment for PL depends on the degree of lameness and grade of PL, the chronicity, and the findings during surgery. In general, during surgery the goals are to achieve good seating of the patella within the femoro-patellar joint and anatomical alignment of the patellar tendon with the direction of quadriceps muscle pull. The surgical steps necessary to achieve these goals differ on a case to case basis.

Avulsion injuries to the tibial tubercle have been described for young racing animals. The growth plates in these patients may not be closed when heavy training is initiated. Chronic strain from the patellar tendon to the cranial tibial tuberosity may lead to avulsion injuries of that structure. Treatment depends on the severity of the injury.⁹ Further possible injuries of the stifle include collateral ligament damage, caudal cruciate ligament damage, fractures of the distal femur, the patella, the proximal tibia or fibula, or avulsion injuries of the cranio-proximal tibial tuberosity in young dogs. Most of these injuries need to be addressed by surgical stabilization and placement of screws, prosthetic ligaments, or bone plates. The prognosis depends on the degree of injury. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities.

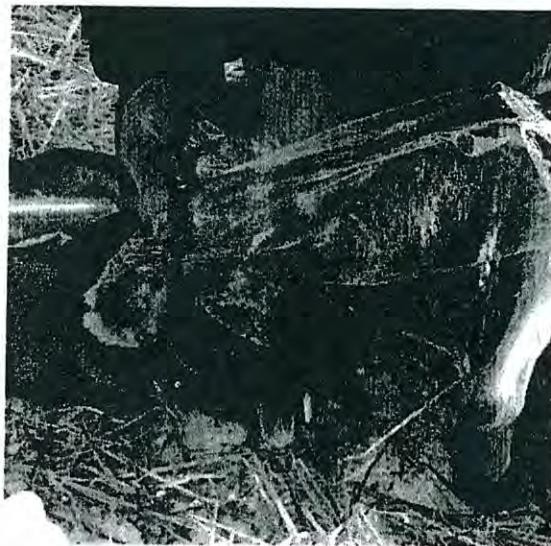
ACHILLES TENDON INJURIES

The diagnosis of Achilles tendon injuries is made by an appropriate orthopedic examination (see previous chapter). Achilles tendon injuries can be long-lasting problems. A dog with an Achilles tendon injury should be dropped. These dogs do not “warm out of the injury”. In contrast, continued exercise will worsen the condition. While weight bearing is possible at 6-8 weeks after injury, complete healing of a tendon may take as long as one year. Tendons never regain 100% of original strength after injury.⁵ If surgical repair is necessary, additional stabilization is achieved by application of a cast, or external fixator. Duration of post – operative care is usually 6-12 weeks. Dogs with chronic Achilles tendon injuries have a guarded prognosis for complete recovery. In contrast, acutely injured animals seem to have a better outcome.¹⁰ In general, the earlier treatment is performed, the better the prognosis. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities. Ultrasound treatment (pulsed 3.3 MHz) of Achilles tendon injuries has been shown to improve results through stimulation of collagen repair. Gastrocnemius muscle tears have a better outcome than tendon injuries. Many will heal with 6-12 weeks of strict confinement and appropriate rehabilitation when healed. Some dogs benefit during the initial weeks of healing from a supportive splint.

TARSAL INJURIES

The diagnosis of tarsal injuries is made by an appropriate orthopedic examination (see previous chapter). Differentials for tarsal injuries include: Ligamentous injuries, fracture, luxation, immune-mediated arthritis, osteochondrosis dissecans, or infection. Similar to carpal injuries, tarsal swelling may be treated successfully with tarsal wraps (Figure 15). For more severe injuries or diseases there is no treatment that can be performed successfully on the trail. With severe instability it may be helpful to apply a splint in an appropriate manner. Depending on the specific injury, surgical repair may allow the animal to return to racing activity. A quick return to full activity may be achieved by using appropriate physical rehabilitation modalities post surgery. With some tarsal injuries however, the prognosis for competitive racing activity is guarded.

Figure 15: Tarsal wraps may improve swelling. Cellophane band is applied to cover the liniment, and vetrap is used to wrap the tarsus.



NEUROLOGIC INJURIES

Injuries to the neck can result from acute trauma such as a fight or severe pull on a neck line. Soft tissue injuries such as a muscle strain may respond to massage therapy, rest, and the administration of a muscle relaxant, such as methocarbamol, and / or a non-steroidal anti-inflammatory. Primary spinal cord injury may be treated using steroids. It is contraindicated to administer NSAIDs and steroids together or within a short period of time. This can lead to significant damage to the gastro intestinal tract, the kidneys, liver or other organ systems. If there is severe, non-ambulatory lameness due to a compression of the spinal canal, for example from a protruded disc that has been diagnosed and localized appropriately via myelogram (Figure 16), computer tomography (CT), or magnetic resonance imaging (MRI), specific surgical procedures are indicated.



Figure 16: A myelogram can help to localize a lesion of the spinal cord. The dog in this figure suffered from intervertebral disc disease (white arrow head).

The location of the lesion often dictates the procedure. A ventral slot may be recommended for treatment of compression of the spinal cord in the neck area. A hemilaminectomy is recommended for treatment of spinal cord compression in the lumbar area. Depending on the underlying cause, the surgical approach and treatment method may differ. Fractures or luxations of the vertebral bodies may be treated using internal fixation methods. It is highly recommended to have a highly trained and experienced small animal surgeon perform these procedures in order to increase chances for an acceptable outcome. Recovery is improved when surgery is followed by a specifically designed protocol under the guidance of a compassionate veterinarian trained in veterinary physical rehabilitation. The prognosis depends on the severity of trauma, the time since injury, and the preoperative condition and neurological status of the animal. Various exercises, especially the use of hydrotherapy during the postoperative period, are likely to help the animal to improve steadily during the recovery period.

Acknowledgements: The authors thank Nelson Priddy II, DVM, DACVS and Mike Edwards, DVM, MS, DACVS for editorial work, and Kelcie Glick for drawing Figure 1.

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